



Tendon-driven soft grasper with asymmetric actuation through a jamming mechanism

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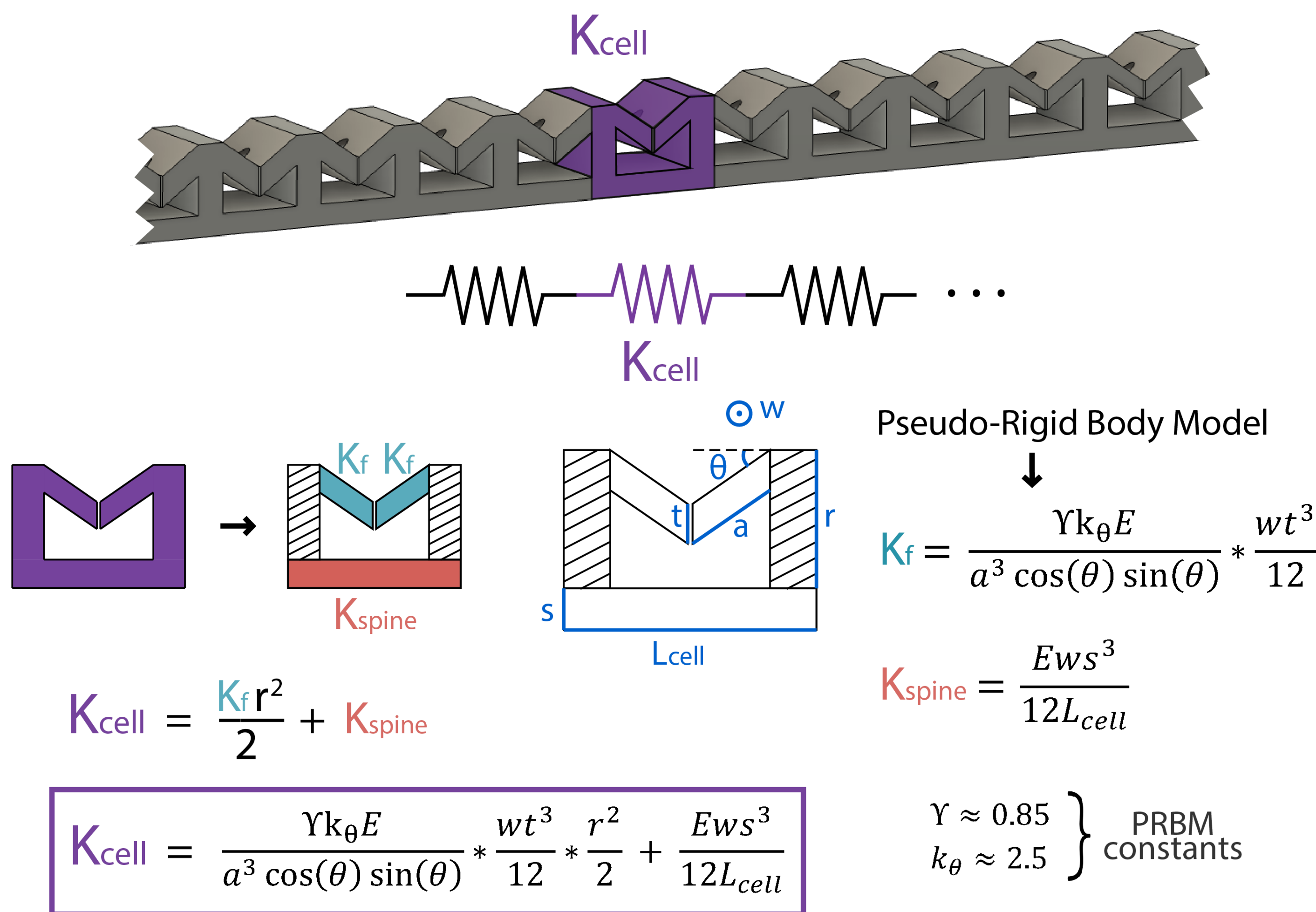


Background

Soft, underactuated graspers enable adaptive manipulation but are often limited to a single actuation profile. Here, programmable stiffness via jamming flexures enable two actuation profiles (sequential and phalangeal), improving conformability and robustness.

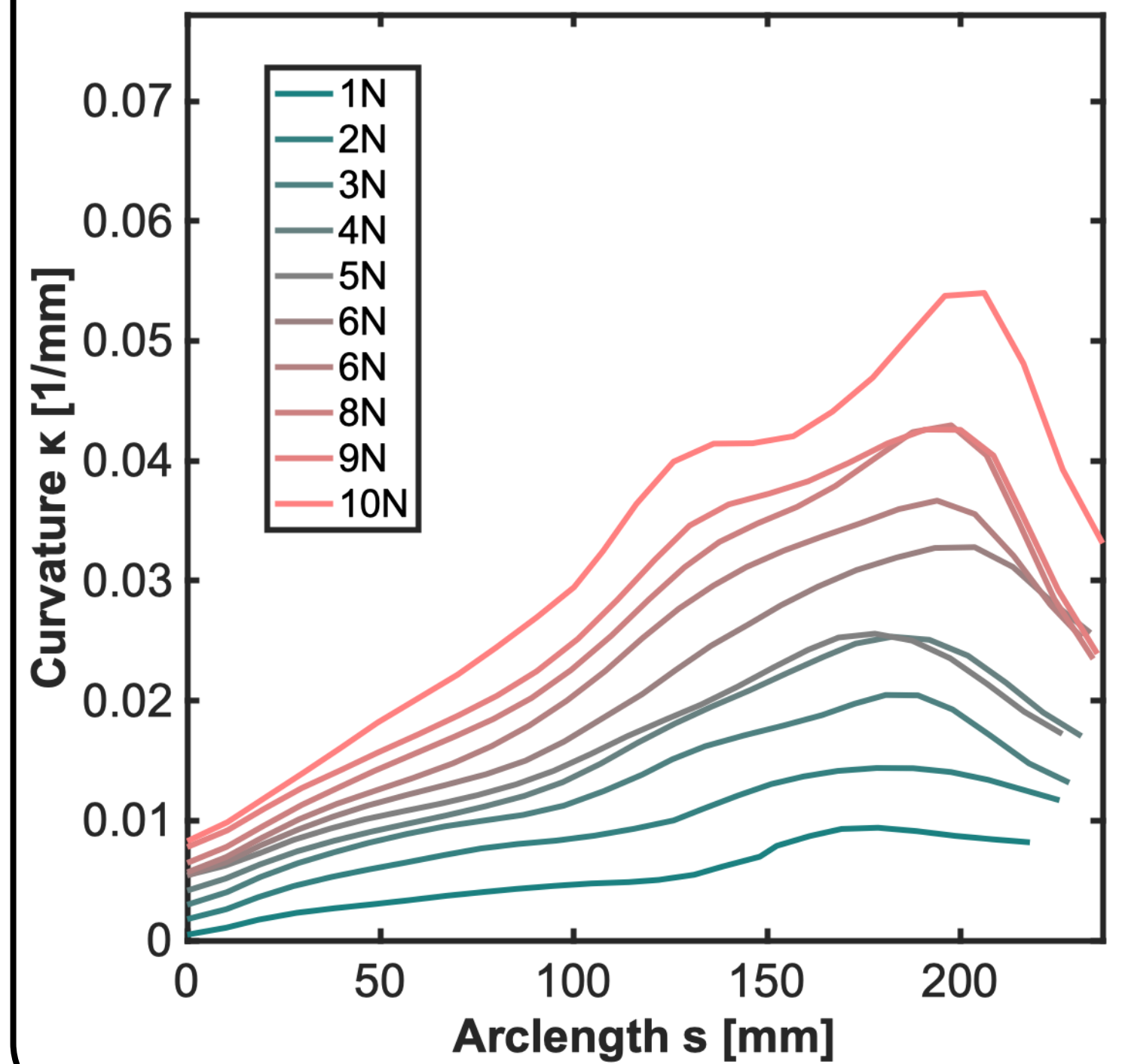
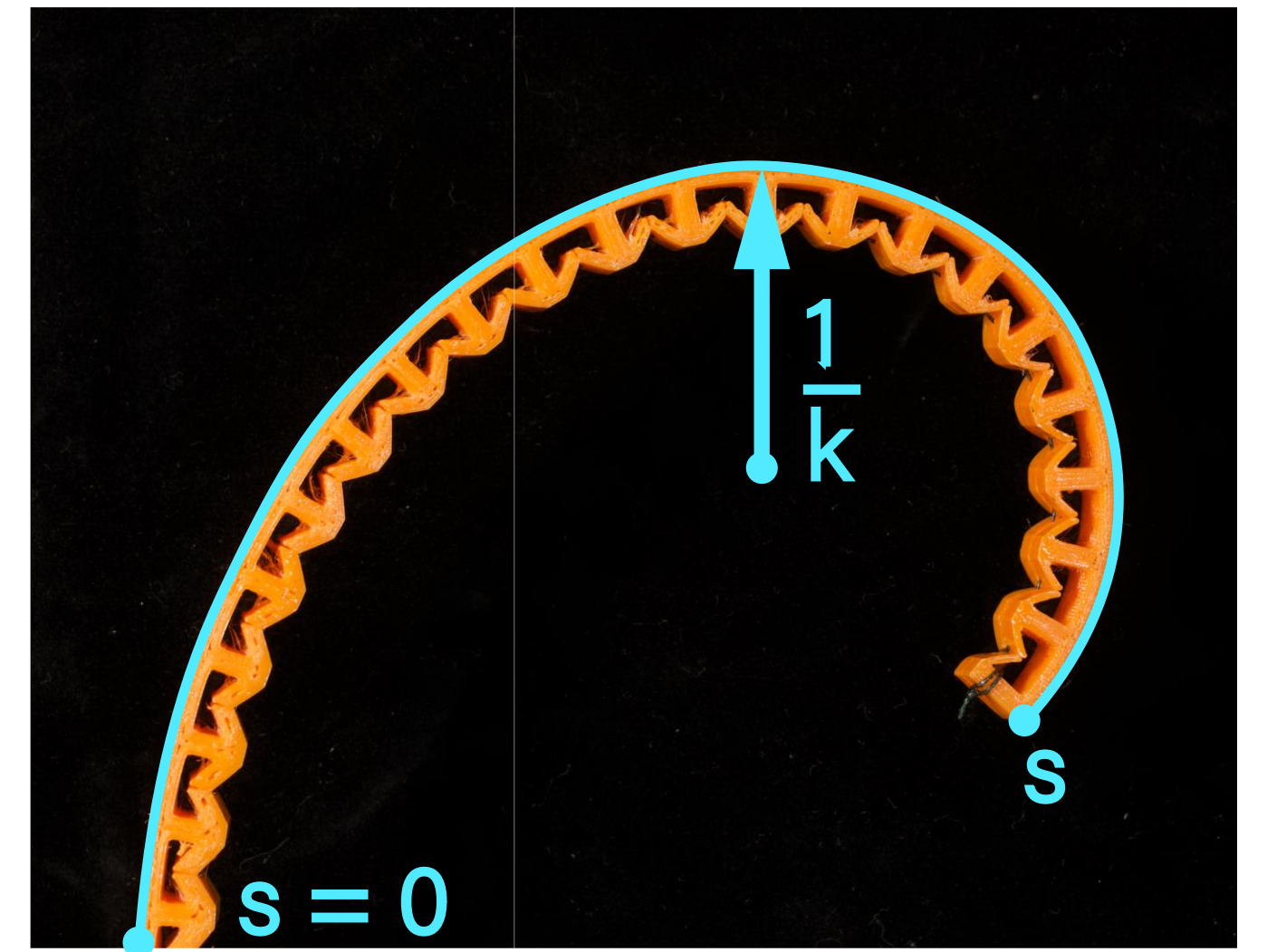
Design Process

- A Pseudo-Rigid Body Model (PRBM) was employed as a design tool to utilize geometric parameters to program the curling profile.
- The design consists of modular, repeating cells of soft flexures that jam together, creating anisotropic stiffness.



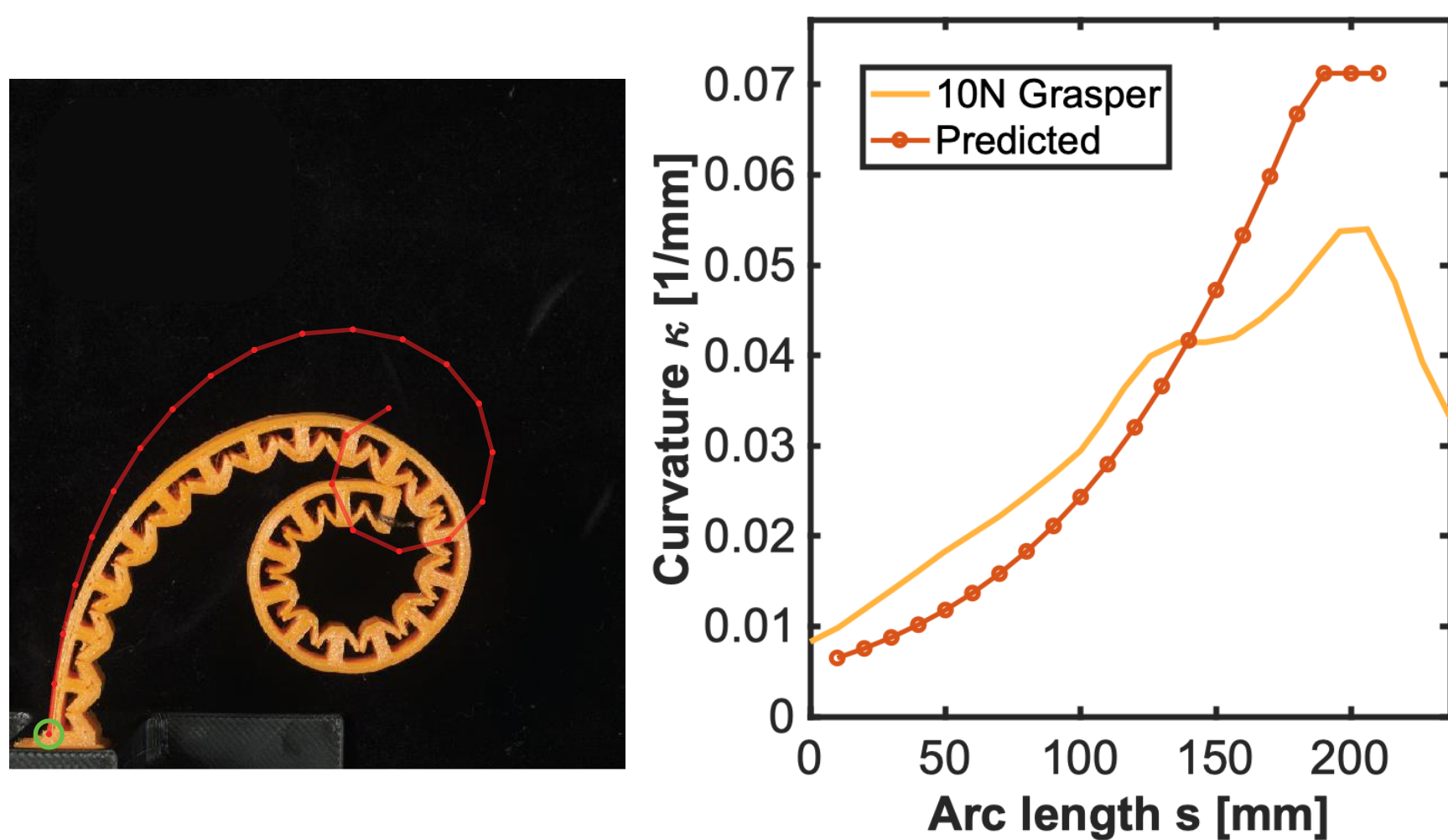
Characterization

A monolithic, FFF 3D-printed, sequentially-actuated, tendon-driven grasper was actuated by pulling the tendon with 1N to 10N of force. Its curvature vs. arclength is shown.



Comparison to Model

The programmable stiffness profile model was compared to the actual curvature profile of the grasper, with 10N applied to the tendon shown.



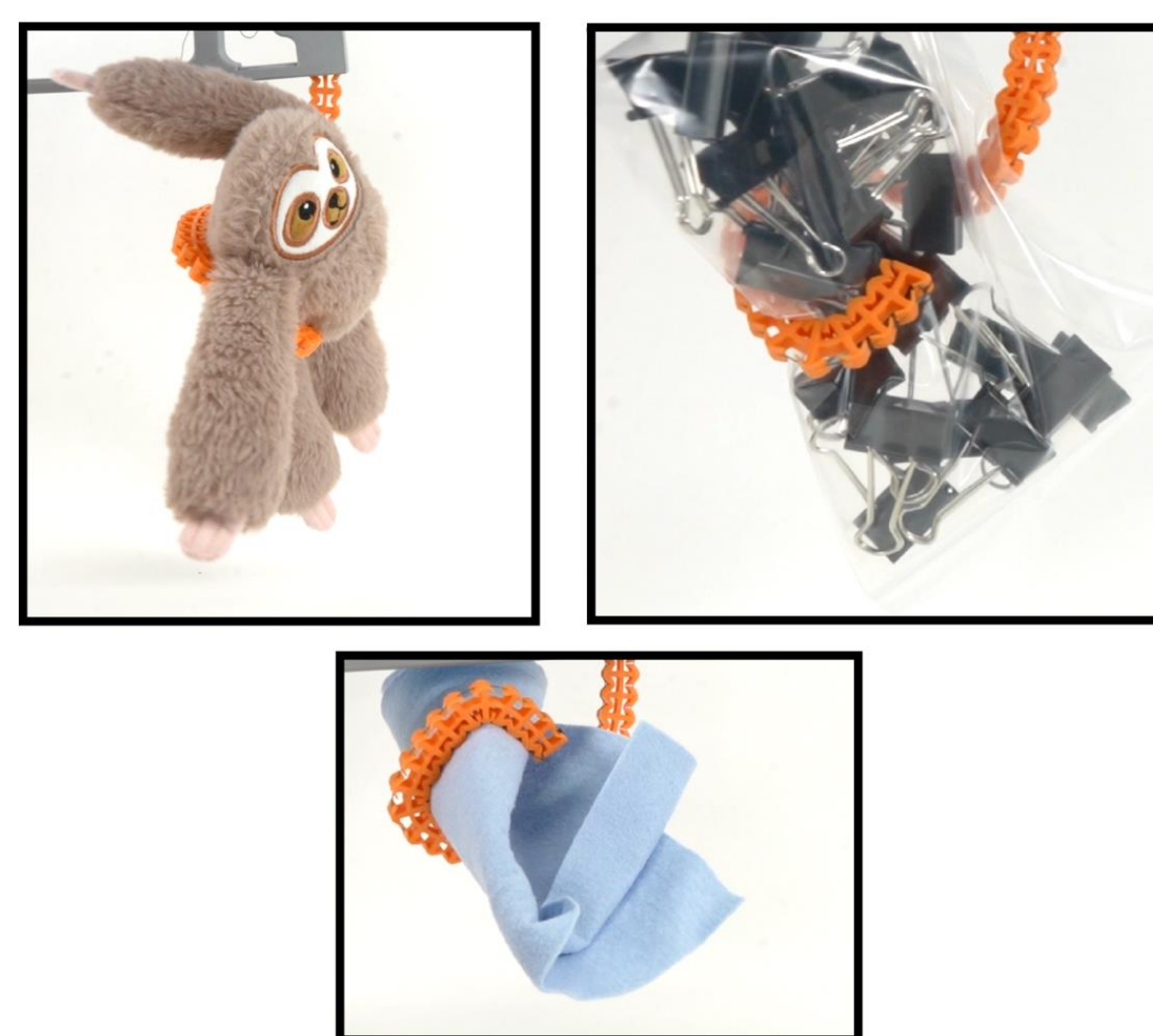
Conclusion

Achieved two distinct curling profiles without needing to reconfigure, enabling programmable and independent stiffness profiles and expanding grasping capability.

Demonstrations

Two independent modes of grasping with the same grasper can enable manipulation of different radii objects

"Phalangeal" Actuation



"Sequential" Actuation

